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The Risk and Return of Equity and Credit Index Options

Contents

1 Big picture

- **Question.** Can a single structural credit-risk model jointly price equity indices, credit indices, SPX options, and CDX options, and can it explain the large returns on these options without implying mispricing?
- **Core setting.** The paper studies SPX equity index derivatives and CDX North American Investment Grade credit index derivatives. CDX options are economically options on credit/debt claims, while SPX options are options on equity claims.
- **Main contribution.** The authors build a structural model in which index-level equity, credit spreads, and options are all priced from the asset dynamics of index constituents.
- **Model ingredients.** Firm asset values have systematic asset shocks, stochastic asset variance, systematic jumps, and idiosyncratic default shocks. The stochastic discount factor prices asset growth risk, variance risk, and jump risk.
- **Estimation strategy.** The benchmark model uses a large homogeneous pool approximation. Only two state variables must be filtered: the representative firm's relative default barrier and systematic asset variance.
- **Inputs for estimation.** The model is estimated using the time series of CDX spread term structures and physical SPX volatility, not option prices. SPX and CDX option prices are therefore out-of-sample tests.
- **Main pricing result.** The model explains SPX and CDX implied volatilities reasonably well out-of-sample, including their volatility skews. It does not imply a systematic inconsistency between equity and credit option markets.
- **Contrast with prior evidence.** The paper revisits the relative-pricing puzzle in Chen et al. and argues that apparent CDX underpricing depends heavily on parameterization, especially the default threshold.
- **Economic mechanism.** Asset risk, variance risk, and jump risk affect equity options and credit options differently. Properly decomposing these three risks is essential to comparing the two markets.
- **Return result.** Average option returns are extreme, but the model-implied expected returns and finite-sample distributions do not reject the model. Large returns can be compensation for systematic risks rather than mispricing.
- **Takeaway.** Equity and credit index option markets look much more consistent once equity, debt, volatility, and jump compensation are priced within one structural framework.

2 Data and measurement

2.1 Markets and instruments

- The equity side is represented by the S&P 500 Index (SPX), SPX physical volatility, and SPX options.
- The credit side is represented by the CDX North American Investment Grade Index, CDX spread term structures, and CDX options.
- CDX options are swaptions written on a broad credit default swap index. A CDX call gains when spreads rise; a CDX put gains when spreads fall.
- SPX puts and CDX calls often pay in similar bad states: equity falls and credit spreads rise.

Interpretation: The paper's distinctive contribution is to treat equity options and credit options as claims written on the same underlying corporate asset process.

2.2 Estimation sample

- The benchmark estimation sample runs from June 2004 to November 2020.
- The estimation inputs are CDX 3-, 5-, 7-, and 10-year spreads plus the conditional physical SPX volatility.
- SPX option tests use SPX implied volatilities over the same long sample.
- CDX option tests begin later, from March 2012 to November 2020, because CDX options only became actively traded in significant volume from 2012 onward.

Interpretation: The design separates estimation from evaluation. The model is disciplined by credit spreads and physical equity volatility, and then judged by how well it prices option markets.

2.3 State variables

The large homogeneous pool approximation leaves two state variables:

- A_D/A_t^r: the relative default barrier of the representative firm;
- V_t: systematic asset variance.

Interpretation: These two states control the level of default risk and the volatility of the common asset factor. They are sufficient for the benchmark model to price SPX, CDX, and their options.

2.4 Option-implied volatility measures

- SPX option prices are converted into Black–Scholes implied volatilities.
- CDX option prices are converted into Black implied volatilities.
- The paper focuses on time-series fit for at-the-money implied volatility and average volatility skews across moneyness.
- SPX moneyness values are typically 95%, 100%, and 105%; CDX option moneyness values are 85%, 100%, and 115%.

Interpretation: Comparing implied volatilities rather than raw option prices makes model and data easier to compare across option maturities and moneyness buckets.

2.5 Option return measures

- The paper constructs returns on quoted calls and puts across moneyness and maturity buckets.
- Model expected excess returns are calculated from instantaneous risk exposures at average filtered states.
- Data counterparts are realized average excess returns from option portfolios.
- The model is judged using p-values from the model-implied finite-sample distribution.

Interpretation: Option returns are noisy and extreme. The paper therefore asks whether realized returns are statistically inconsistent with model-implied risk premia, not whether they exactly equal the model mean.

3 Models and empirical specifications

3.1 Firm asset dynamics

For firm j, the asset value dynamics under the physical measure are

$$\frac{dA_t^j}{A_t^j} = \frac{dA_t^m}{A_t^m} + \sigma_j dW_t^j + \nu_j dN_t^j - \lambda^j \nu_j dt. \tag{1}$$

The market asset factor follows

$$\frac{dA_t^m}{A_t^m} = (\mu_t - q)dt + \sqrt{V_t} dW_t^{A,m} + \nu_t^m dN_t^m - \lambda_t^m \nu_t^m dt, \tag{2}$$

$$dV_t = \kappa(\theta - V_t)dt + \delta\sqrt{V_t} dW_t^{V,m}. \tag{3}$$

Interpretation: The model separates systematic asset growth risk, systematic variance risk, systematic jump risk, and idiosyncratic default risk.

3.2 Stochastic discount factor

The stochastic discount factor is exponentially affine in aggregate risks:

$$\frac{d\phi_t}{\phi_t} = -r_f dt - \xi_{A,LV}\sqrt{V_t} dW_t^{A,LV} - \xi_V \sqrt{V_t} dW_t^V + \left(e^{\xi_Z^m Z_t^m} - 1\right) dN_t^m - \lambda_t^m \xi_Z^m \nu_t^m dt. \tag{4}$$

Interpretation: This SDF prices three systematic risks: asset growth, variance, and jumps. Idiosyncratic risks are deliberately not priced.

3.3 Risk-neutral dynamics and risk premia

The risk-neutral dynamics imply an unlevered asset risk premium:

$$(\mu_t - r_f)dt = \left(\sqrt{1-\rho^2}\xi_{A,LV} + \rho\xi_V\right) V_t dt + \lambda_t^m \mathbb{E}\left[\left(1 - e^{\xi_Z^m Z_t^m}\right)\left(e^{Z_t^m} - 1\right)\right] dt. \tag{5}$$

Interpretation: The first term prices diffusive asset and variance risk; the second term prices systematic asset jump risk.

3.4 Default and corporate securities

Firm j defaults when its asset value hits the default barrier:

$$\tau_j = \inf\{s \geq t : A_s^j \leq A_D\}. \tag{6}$$

Debt value is

$$D(A_t^j, V_t) = \frac{c}{r_f}\left[1 - P_D(A_t^j, V_t)\right] + (1-\alpha)A_D P_D(A_t^j, V_t), \tag{7}$$

where P_D is the risk-neutral probability of default.

Interpretation: Equity and credit claims are endogenous functions of the same underlying assets. Credit spreads and equity values are therefore tied together by construction.

3.5 Index construction and large homogeneous pool

The benchmark model approximates the index by a large pool of ex ante identical firms. This implies that:

- idiosyncratic firm shocks diversify away in the index;
- the SPX and CDX depend only on the two systematic state variables;
- option pricing can be done with a two-dimensional state space.

Interpretation: The homogeneous pool is a computational device. The paper later relaxes it using industry subindices to check whether heterogeneity matters.

3.6 Maximum likelihood filtering

The state variables are filtered by matching the 5-year CDX spread and physical SPX volatility:

$$S_{5,t}^{CDX} = S(A_t^r, V_t, 5, \Theta), \qquad \sigma_t^{SPX} = \sqrt{\sigma_E^2(A_t^r, V_t, \Theta)}. \tag{8}$$

The remaining CDX spreads are treated as noisy observations in the likelihood:

$$\frac{S_{T,t}^{CDX} - S(A_t^r, V_t, T, \Theta)}{S_{T,t}^{CDX}} = e_{T,t}, \qquad T = 3, 7, 10. \tag{9}$$

The parameter vector is estimated by

$$\widehat{\Theta} = \arg\max_{\Theta} \mathcal{L}(\Theta). \tag{10}$$

Interpretation: The procedure uses time-series restrictions on state variables and avoids fitting each option market independently.

4 Identification and empirical design

4.1 What identifies the model

- CDX spread term structures identify credit-risk levels and default-related dynamics.
- Physical SPX volatility identifies systematic variance.
- The time-series likelihood disciplines the evolution of filtered state variables.
- SPX and CDX option prices are not used in estimation, so their fit is out-of-sample.

Interpretation: The model is identified by joint restrictions across credit spreads, physical volatility, and time-series dynamics, not by freely fitting implied volatilities.

4.2 Why the comparison with CDJT matters

- Prior work argued that SPX and CDX options are hard to reconcile: CDX options seemed underpriced relative to SPX options.
- This paper shows that the puzzle appears under CDJT-style parameterization but disappears under the authors' estimated parameterization.
- The key difference is the default threshold. CDJT use a threshold based on total debt; this paper's estimate is closer to total liabilities.

Interpretation: A structural model's relative-pricing conclusion can be highly sensitive to leverage and default-threshold calibration.

4.3 Why option returns are a separate test

- Pricing levels can match while risk premia remain implausible.
- The paper therefore examines realized average excess returns on SPX and CDX options.
- It compares realized returns to model-implied expected returns and model-implied finite-sample distributions.

Interpretation: The return test asks whether option markets are overpriced or underpriced after accounting for the risk exposures generated by the model.

4.4 Limitations

- The homogeneous-pool approximation abstracts from cross-sectional firm heterogeneity.
- CDX option data are available only from 2012 onward.
- Option returns are extremely noisy, reducing statistical power.
- The model still underprices some out-of-the-money SPX puts and CDX calls, though not enough to imply relative mispricing.

Interpretation: The paper's claim is not perfect option pricing. It claims that a coherent structural model can jointly rationalize levels and returns much better than prior relative-pricing evidence suggested.

5 Tables and figures

5.1 Tables 1–2: structural parameters and nested risk tests

Read:

- Table 1 reports calibrated parameters and maximum-likelihood estimates. The default barrier is 19.50, and the estimated long-run mean of variance implies roughly 9.41% annual asset volatility.
- The estimated market price of variance risk is negative, and systematic jump-risk parameters imply an important jump component of total asset risk compensation.
- Table 2 shows that removing systematic variance risk or systematic jump risk sharply lowers the log-likelihood. The likelihood-ratio p-values are essentially zero.

Economic message: Variance risk and jump risk are not optional embellishments. The data reject models without them.

Structural parameters. The table reports the calibrated and estimated values for the structural model parameters. Panel A presents the parameters that are calibrated, and Panel B presents the parameters that are obtained via maximum likelihood estimation. Reported together in square brackets are robust standard errors calculated using the Huber sandwich estimator. The standard errors are expressed in the same units as the corresponding parameter estimates. The estimation sample period is from June 2004 to November 2020.

(A) Calibrated parameters		
1-Distress costs (α)	25.00%	
2-Corporate tax rate (ζ)	20.00%	
3-Recovery rate (R)	51.00%	
4-Risk-free interest rate (r_f)	1.00%	
5-Asset payout rate (q)	2.00%	
6-Coupon (c)	0.25	
7-Systematic jump intensity loading (η_m: $\lambda_t^m = \eta_m V_t$)	112.92	
8-Idiosyncratic jump intensity (λ^j)	1.00%	
9-Default barrier (A_D)	19.50	
(B) Estimated parameters		
1-Mean reversion speed ($\hat{\kappa}$)	2.6637	[0.0584]
2-Long run mean ($\hat{\theta}$)	0.0089	[0.0004]
3-Volatility parameter for asset variance ($\hat{\delta}$)	21.49%	[1.5191]
4-Correlation between asset value and variance shocks ($\hat{\rho}$)	−0.6004	[0.0635]
5-Market price of asset specific risk ($\hat{\xi}_{A\perp V}$)	0.4759	[16.822]
6-Market price of variance risk ($\hat{\xi}_V$)	−6.1029	[0.4439]
7-Systematic jump size mean ($\hat{z}_m$)	−3.24%	[0.4992]
8-Systematic jump size standard deviation ($\hat{\gamma}_m$)	1.17%	[0.1923]
9-Market price of systematic jump risk($\hat{\xi}_m$)	−3.0010	[0.1398]
10-Idiosyncratic volatility ($\hat{\sigma}_j$)	9.08%	[0.5506]

(a) Table 1: structural parameters.

lihood ratio tests. The table performs likelihood ratio tests for two nested versions of our model: ystematic variance risk and no systematic jump risk. For each case, the log-likelihood ratio test (LRT) stic is defined as $2 \times \log\left[\mathcal{L}(\Theta_{\text{full}})/\mathcal{L}(\Theta_{\text{nested}})\right]$, and the p-value is calculated based on the chi-square ibution. The degrees of freedom are determined to be the number of model restrictions.

	(A) Full model	(B) Nested models	
		No systematic variance risk	No systematic jump risk
Log-likelihood	46197.57	41510.48	45630.07
LRT statistic		9374.19	1135.01
egrees of freedom		4	4
hi-square p-value		0.00	0.00

(b) Table 2: likelihood-ratio tests.

5.2 Table 3 + Figure 1: model mechanisms and risk exposures

Read:

- Table 3 shows how CDX spreads, SPX implied volatility, and CDX implied volatility vary with the relative default barrier and asset variance.
- CDX spreads rise with both the default barrier and asset variance.
- SPX implied volatility rises monotonically with asset variance.
- CDX implied volatility can display hump-shaped patterns because higher default risk and higher leverage pull in different directions.
- Figure 1 decomposes exposures to asset, variance, and jump risks for SPX/CDX indices and options.

Economic message: The same shocks affect equity and credit derivatives differently. This is why comparing SPX and CDX option prices requires a structural risk-decomposition framework.

Model results as functions of the state variables. The table reports the 5-year CDX spread (Panel A) and 1-month SPX and CDX implied volatilities (Panels B and C) from the model. We evaluate the model at the following nine combinations of the two state variables: $(\frac{A_D}{A_t^r}, V_t) \in \{0.35022, 0.38244, 0.49218\} \times \{0.00097, 0.00619, 0.02028\}$. The CDX spread is expressed in basis points. The implied volatilities are expressed in percentages and are calculated at 95, 100, and 105% moneyness values for SPX options and at 85, 100, and 115% for CDX options. Since we evaluate the model at extreme state values in this exercise, we increase the number of simulations to 100,000 paths ×500 firms for accuracy.

		(A) CDX	(B) SPX options			(C) CDX options		
			95%	100%	105%	85%	100%	115%
		Low relative default barrier A_D/A_t^r						
V_t	10th	41.65	10.12	6.52	5.78	18.41	25.80	35.67
	Median	43.78	15.50	13.10	11.20	41.98	50.98	57.46
	90th	50.67	24.52	22.92	21.50	82.83	84.82	86.61
		Mid relative default barrier A_D/A_t^r						
V_t	10th	58.38	10.49	6.86	6.01	17.26	27.49	35.97
	Median	64.36	16.21	13.79	11.84	41.42	48.18	52.06
	90th	73.71	25.75	24.13	22.67	71.52	71.62	72.81
		High relative default barrier A_D/A_t^r						
V_t	10th	141.38	12.06	8.32	7.08	14.51	19.59	27.79
	Median	148.64	19.27	16.76	14.64	28.05	34.95	41.11

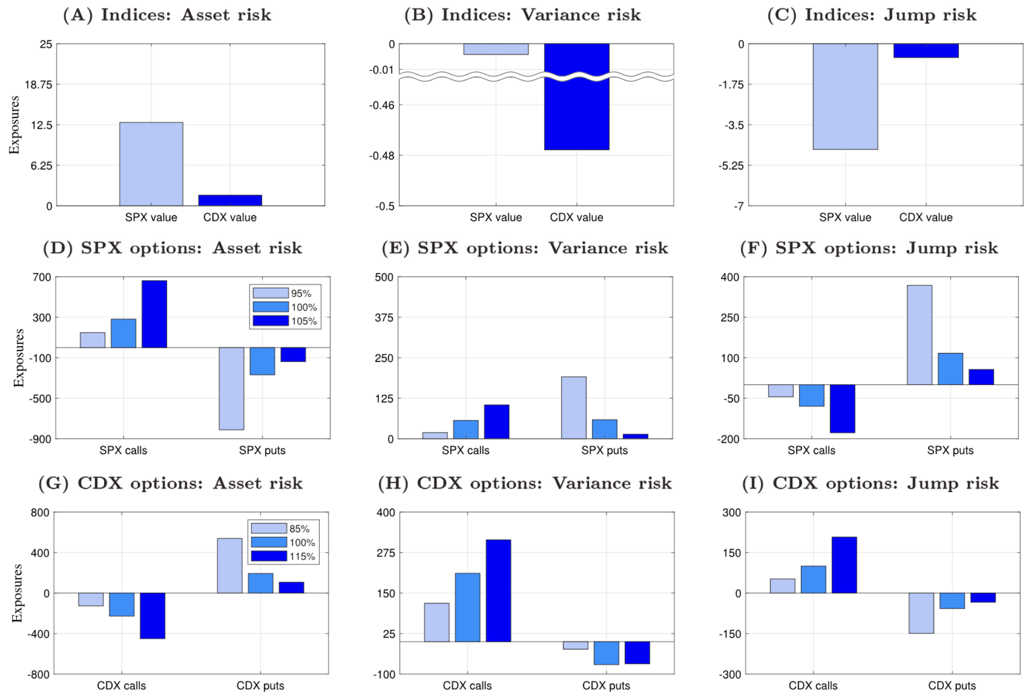


Fig. 1. Exposures to three sources of systematic risk. The figure graphs the three exposures implied by the model for the SPX/CDX (Panels A, B, and C), for SPX options (Panels D, E, and F), and for CDX options (Panels G, H, and I). We consider 1-month calls and puts with moneyness of 95, 100, and 105% for SPX options and 85, 100, and 115% for CDX options. To facilitate comparison, we normalize the exposures so that they capture the response to a one standard deviation change in one of the three sources of systematic risk. Specifically, the left panels plot $\Delta_{A,t}\sqrt{V_t}$, the center panels plot $\Delta_{V,t}\delta\sqrt{V_t}$, and the right panels plot $\bar{\Delta}_{N,t}\sqrt{\lambda_t^m}$. The results are produced by setting the two state variables at $(A_D/A_t^r, V_t) = (0.38244, 0.00619)$.

5.3 Table 4: pricing consistency and the CDJT puzzle

Read:

- Table 4 reproduces the CDJT-style underpricing result when their parameterization and state-variable choices are imposed.
- Under the authors' benchmark parameterization, CDX implied volatility is close to or slightly higher than its data counterpart.
- Adjusting the default threshold toward CDJT's value is the main change that recreates the underpricing puzzle.

Economic message: The apparent inconsistency between equity and credit option markets is not robust to a time-series-consistent structural estimation of default thresholds and state variables.

Pricing consistency. The table revisits the relative pricing puzzle posed by CDJT based on our model. Panel A reports the data median for the CDX spread as well as for the at-the-money SPX and CDX implied volatilities. The sample period is from March 2012 to November 2020, during which the data on both options are available. The other two panels calculate the CDX implied volatility from the model when the two state variables are chosen to exactly fit the CDX spread and the SPX implied volatility from the data under our benchmark parametrization (Panel B) and under CDJT's parametrization (Panel C). Reported together are three key model inputs. To understand why the two parametrizations produce different results, in Panel B, we begin with our benchmark parametrization and gradually change the values of the parameters and state variables until we reach CDJT's parametrization.

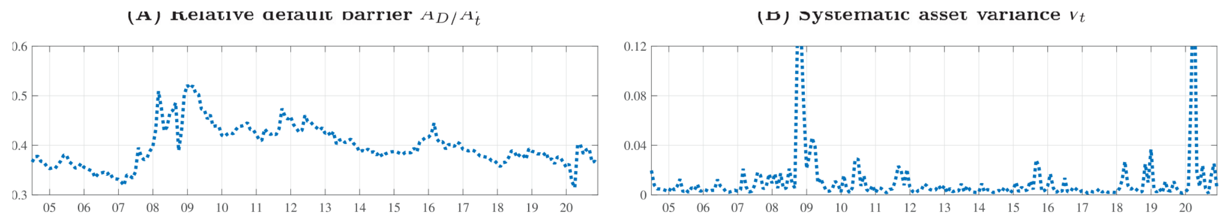
	Target	Implied volatility		Model inputs		
	CDX	SPX	CDX	A_D/A_t^r	V_t	σ_j
(A) Data median (Mar 2012 - Nov 2020)	67.50	13.27	43.39			
(B) Our benchmark parametrization	67.50	13.27	46.35	0.3880	0.0056	0.0980
+ (1) Lower A_D/A_t^r but higher σ_j	67.50	9.90	25.81	0.1712	0.0056	0.3204
+ (2) Higher asset variance V_t	67.50	13.42	32.62	0.1712	0.0108	0.3178
+ (3) Other changes to reach CDJT	67.50	13.27	33.44	0.1712	0.0108	0.2840
(C) CDJT parametrization	67.50	13.27	33.44	0.1712	0.0108	0.2840

5.4 Figures 2–3 + Table 5: filtered states and in-sample fit

Read:

- Figure 2 plots the filtered relative default barrier and systematic asset variance. Variance spikes during the Global Financial Crisis and COVID-19.
- Figure 3 shows that the model tracks 3-, 5-, 7-, and 10-year CDX spreads and physical SPX volatility well in-sample.
- Table 5 reports high correlations between model and data for CDX spreads, with correlations around 0.86–0.97.

Economic message: The state variables are economically sensible and the model fits the inputs used for estimation without generating implausible time-series behavior.



. 2. **Filtered state variables.** The figure presents the time series of the two filtered state variables: the relative default barrier $A_D/\hat{A}_t^r$ (Panel A) and systematic asset variance (Panel B). The measurement frequency is monthly, where the values of the two filtered state variables are sampled at the end of each month. The sample period is from June

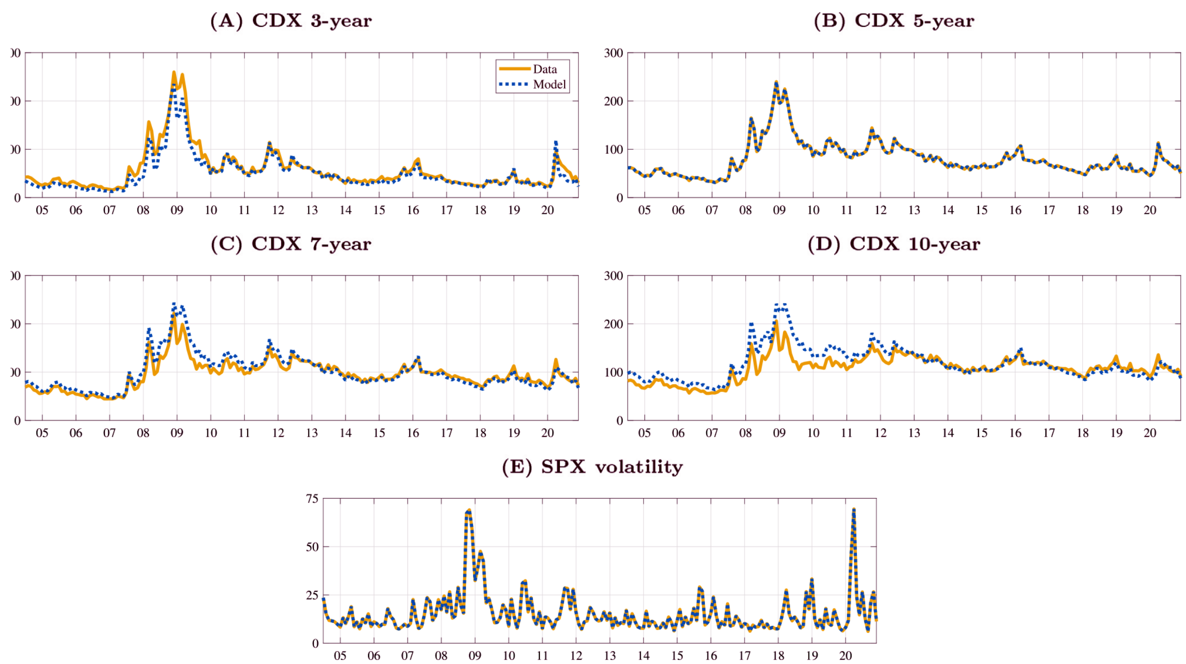


Table 5

In-sample model goodness of fit. The table examines the in-sample model goodness of fit. Panels A and B report the sample means and standard deviations of the 3-, 5-, 7-, and 10-year CDX spreads and the physical SPX volatility in the data and in the model. In Panel C, we present two goodness of fit metrics to assess model performance: the time-series correlation between the data and the model as well as the root mean squared error (RMSE). The measurement frequency is monthly, where the data and model values are sampled at the end of each month. The sample period is from June 2004 to November 2020.

	(A) Data		(B) Model		(C) Fit	
	Mean	Std.	Mean	Std.	Corr.	RMSE
CDX 3-year	56.25	42.66	48.84	36.31	0.97	13.23
CDX 5-year	79.85	35.77	79.85	35.77	–	–
CDX 7-year	94.93	29.62	99.83	36.48	0.96	12.68
CDX 10-year	108.45	25.95	118.37	33.91	0.86	20.11
SPX volatility	15.46	10.40	15.46	10.40	–	–

5.5 Figure 4 + Table 6: SPX option pricing

Read:

- Figure 4 compares SPX implied volatilities in data and model. The model tracks the level and major time-series patterns but misses some extreme spikes.
- Table 6 reports out-of-sample SPX option fit. Correlations between model and data are about 0.84–0.87, and RMSEs are moderate.
- The model captures the average SPX volatility skew, though its skew is slightly flatter than in the data.

Economic message: A model estimated without option prices can still capture much of the SPX option surface, suggesting that structural asset and variance dynamics carry substantial option-pricing information.

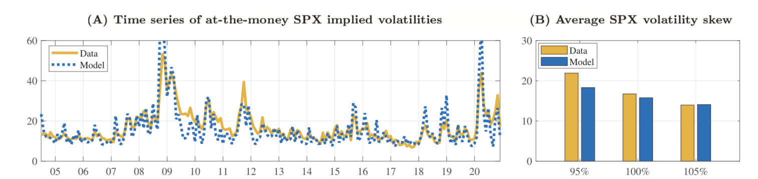


Fig. 4. SPX implied volatilities. The figure compares the Black–Scholes-implied volatilities for SPX options in the data and in the model. Panel A plots the time series of at-the-money SPX implied volatilities in the data (yellow solid line) and in the model (blue dotted line). Panel B presents the average SPX implied volatilities with moneyness of 95, 100, and 105% in the data (yellow bars) and in the model (blue bars). The data frequency is monthly, where the data and model values are sampled at the end of each month. The implied volatilities are expressed in percentages. The sample period is from June 2004 to November 2020.

(a) Figure 4: SPX implied volatilities.

Table 6

Out-of-sample model fit for SPX options. The table examines the out-of-sample fit for SPX options. Panels A and B report the sample means and standard deviations of Black–Scholes-implied volatilities for SPX options with moneyness of 95, 100, and 105% in the data and in the model. In Panel C, we present two goodness of fit metrics to assess model performance: the time-series correlation between the data and the model as well as the root mean squared error (RMSE). The measurement frequency is monthly, where the data and model values are sampled at the end of each month. The sample period is from June 2004 to November 2020.

	(A) Data		(B) Model		(C) Fit	
K/I_t	Mean	Std.	Mean	Std.	Corr.	RMSE
95%	21.91	7.55	18.32	9.56	0.84	6.29
100%	16.74	7.87	15.79	9.99	0.87	5.14
105%	13.96	7.15	14.08	10.03	0.86	5.30

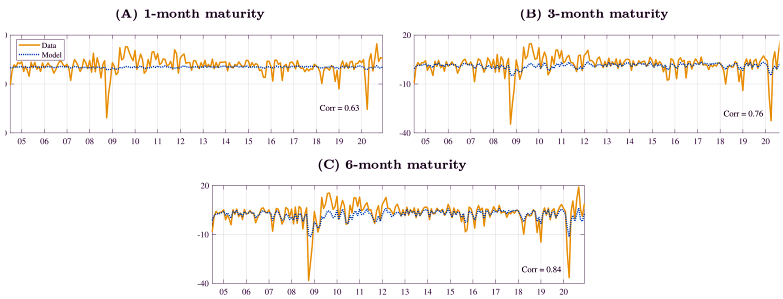
(b) Table 6: SPX option fit.

5.6 Figure 5 + Table 7: volatility risk premium and CDX option pricing

Read:

- Figure 5 plots the volatility risk premium, defined as the difference between at-the-money SPX implied volatility and physical SPX volatility.
- The model captures the broad level but the volatility-risk-premium series is flatter than in the data at short horizons.
- Table 7 reports CDX option performance. Correlations between data and model implied volatilities are about 0.74–0.78.

Economic message: The model performs reasonably for a newer, less liquid market. It captures CDX option levels and skews without implying systematic CDX underpricing.



(a) Figure 5: volatility risk premium.

Table 7
Out-of-sample model fit for CDX options. The table examines the out-of-sample fit for CDX options. Panels A and B report the sample means and standard deviations of Black-implied volatilities for CDX options with moneyness of 85, 100, and 115% in the data and in the model. In Panel C, we present two goodness of fit metrics to assess model performance: the time-series correlation between the data and the model as well as the root mean squared error (RMSE). The measurement frequency is monthly, where the data and model values are sampled at the end of each month. The sample period is from March 2012 to November 2020.

	(A) Data		(B) Model		(C) Fit	
K/I_t	Mean	Std.	Mean	Std.	Corr.	RMSE
85%	39.75	15.16	37.72	22.10	0.74	14.92
100%	46.74	13.99	45.49	18.20	0.77	11.63
115%	55.02	13.10	50.20	16.72	0.78	11.39

(b) Table 7: CDX option fit.

5.7 Figures 6–8: CDX volatility, industry heterogeneity, and homogeneity check

Read:

- Figure 6 shows CDX implied volatility levels and skews. The model captures the average positive CDX volatility skew.
- Figure 7 introduces five CDX industry subindices and shows substantial heterogeneity, especially for financials during the crisis.
- Figure 8 compares implied-volatility skews under firm heterogeneity and homogeneity. The differences are modest.

Economic message: Firm heterogeneity matters, but the benchmark homogeneous model's conclusions about option pricing are fairly robust.

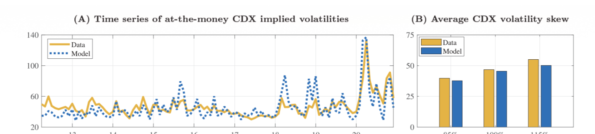


Fig. 6. CDX implied volatilities. The figure compares the Black-implied volatilities for CDX options in the data and in the model. Panel A plots the time series of at-the-money CDX implied volatilities in the data (yellow solid line) and in the model (blue dotted line). Panel B presents the average CDX implied volatilities with moneyness of 85, 100, and 115% in the data (yellow bars) and in the model (blue bars). The data frequency is monthly, where the data and model values are sampled at the end of each month. The implied volatilities are expressed in percentages. The sample period is from March 2012 to November 2020.

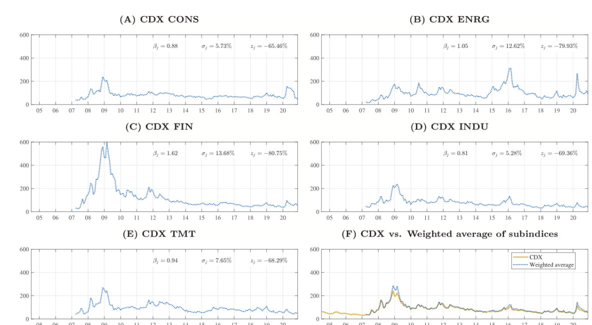


Fig. 7. CDX industry subindices. The figure plots the monthly time series of the 5-year CDX subindex spreads. There are five industry subindices: consumer cyclical (CONS, Panel A); energy (ENRG, Panel B); financials (FIN, Panel C); industrial (INDU, Panel D); telecom, media, and technology (TMT, Panel E). In addition, Panel F calculates the weighted average spread among the five subindices (blue dotted line) and compares it with the CDX spread (yellow solid line). The spreads are expressed in basis points. The sample period is from March 2007 to November 2020.

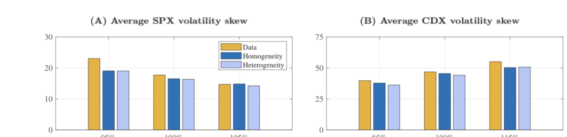


Fig. 8. Firm homogeneity vs. heterogeneity. The figure juxtaposes the average implied volatility skews under heterogeneity (lighter blue bars) and those under homogeneity (darker blue bars), along with the data (yellow bars). Panel A presents the average SPX implied volatilities with moneyness of 95, 100, and 105%, and the sample period is from March 2007 to November 2020. Panel B presents the average CDX implied volatilities with moneyness of 85, 100, and 115%, and the sample period is from March 2012 to November 2020. The data frequency is monthly, where the data and model values are sampled at the end of each month. The implied volatilities are expressed in percentages.

5.8 Tables 8–9: option returns and risk-premium decomposition

Read:

- Table 8 compares realized average excess returns with model-implied expected excess returns for SPX and CDX options.
- SPX calls and CDX puts have positive average expected returns; SPX puts and CDX calls have negative average expected returns.

- The model does not reject the realized average option returns at conventional significance levels.
- Table 9 decomposes expected returns into asset, variance, and jump risk premia.
- Asset risk dominates for SPX calls and CDX puts; variance risk is especially important for CDX options; jump risk contributes smaller but economically meaningful shares.

Economic message: Extreme option returns are interpretable as compensation for risk exposures. The model provides no evidence of systematic mispricing between equity and credit options.

Table 8

Average excess returns vs. expected excess returns. The table compares the average excess returns calculated from the data and the expected excess returns implied by the model. Panel A presents the results for SPX options, whereas Panel B presents the results for CDX options. For the model, we calculate the instantaneous expected excess returns on 1-month calls and puts by evaluating Eq. (14) at the average filtered state variables. The moneyness values are 95, 100, and 105% for SPX options and 85, 100, and 115% for CDX options. For the data counterparts, we construct the portfolios of calls and puts based on moneyness and maturity buckets. Each portfolio is formed by collecting quoted calls/puts whose moneyness values are within ±2.5% of the target and whose maturities range from 15 to 60 days. We then calculate the annualized sample mean of daily excess returns on each option portfolio. The *p*-values are calculated by simulating the model-implied finite-sample distribution of the average excess return. All returns are annualized and expressed in percentages. The sample period is from June 2004 to November 2020 for SPX options and from March 2012 to November 2020 for CDX options.

	(A) SPX options			(B) CDX options		
	Data	Model	*p*-value	Data	Model	*p*-value
Call ITM	98.25	89.67	0.90	−174.00	−166.52	0.94
Call ATM	191.55	138.22	0.59	−372.62	−275.11	0.51
Call OTM	170.81	278.76	0.41	−716.07	−440.65	0.13
Put OTM	−462.81	−474.90	0.93	437.60	328.68	0.54
Put ATM	−283.80	−214.92	0.53	161.91	181.57	0.87
Put ITM	−151.57	−115.74	0.64	104.90	124.50	0.82

(a) Table 8: realized vs. expected returns.

Table 9

Risk premium decomposition. The table decomposes the expected excess returns from the model into three risk premia: asset risk premium, variance risk premium, and jump risk premium. Panel A presents the results for the SPX and SPX options, whereas Panel B presents the results for the CDX and CDX options. Reported together in square brackets are percentage shares of individual risk premia to the total risk premium. The decomposition is obtained by evaluating Eq. (14) at the average filtered state variables. All returns are annual and expressed in percentages.

	Total	=	Asset risk	+	Variance risk	+	Jump risk
(A) SPX options							
Call ITM	89.67 [100%]		97.69 [109%]		−15.70 [-18%]		7.68 [9%]
Call ATM	138.22 [100%]		165.38 [120%]		−39.33 [-28%]		12.16 [9%]
Call OTM	278.76 [100%]		330.17 [118%]		−74.34 [-27%]		22.94 [8%]
Put OTM	−474.90 [100%]		−339.10 [71%]		−94.63 [20%]		−41.16 [9%]
Put ATM	−214.92 [100%]		−157.48 [73%]		−39.21 [18%]		−18.23 [8%]
Put ITM	−115.74 [100%]		−92.65 [80%]		−12.93 [11%]		−10.16 [9%]
(B) CDX options							
	Total	=	Asset risk	+	Variance risk	+	Jump risk
Call ITM	−166.52 [100%]		−78.15 [47%]		−79.80 [48%]		−8.57 [5%]
Call ATM	−275.11 [100%]		−128.12 [47%]		−131.96 [48%]		−15.03 [5%]
Call OTM	−440.65 [100%]		−218.88 [50%]		−194.38 [44%]		−27.39 [6%]
Put OTM	328.68 [100%]		269.61 [82%]		39.80 [12%]		19.27 [6%]
Put ATM	181.57 [100%]		118.59 [65%]		53.81 [30%]		9.17 [5%]
Put ITM	124.50 [100%]		70.74 [57%]		47.83 [38%]		5.93 [5%]

(b) Table 9: risk-premium decomposition.

6 Short summary

- The paper develops a structural credit-risk model for equity indices, credit indices, and their options.
- The model prices claims from constituent-firm asset dynamics with stochastic variance and systematic jumps.
- The benchmark large homogeneous pool approximation reduces the state space to relative default barrier and asset variance.
- The model is estimated using CDX spreads and physical SPX volatility, not option prices.
- Likelihood-ratio tests reject models without systematic variance risk or systematic jump risk.
- The model fits in-sample CDX spreads and physical SPX volatility well.
- Out-of-sample, it fits SPX and CDX implied volatilities reasonably well.
- The paper does not find systematic relative mispricing between SPX and CDX options.
- The prior CDX underpricing puzzle depends importantly on the default-threshold parameterization.
- Option returns are extreme, but not statistically inconsistent with model-implied risk premia.
- Asset, variance, and jump risks all matter, but their importance differs across equity and credit derivative contracts.

7 Conclusion

7.1 Core conclusion

Doshi, Ericsson, Fournier, and Seo show that a structural credit-risk model can jointly explain equity index options and credit index options. Once equity, debt, variance, and jump risks are linked through the same constituent-firm asset process, the apparent inconsistency between SPX and CDX option markets largely disappears.

7.2 Economic intuition

Equity and credit derivatives are claims on different parts of the same capital structure. Equity is more exposed to asset-value upside and downside; credit instruments are more sensitive to default probabilities and spread volatility. Options on these claims inherit different exposures to asset, variance, and jump risks. A model that attributes these exposures correctly can rationalize both option prices and option returns.

7.3 Takeaway

The paper's main message is that the relative pricing of equity and credit options cannot be judged from reduced-form implied-volatility comparisons alone. A structural model is needed to map the same underlying risks into equity, credit, and option payoffs. Under that mapping, both option markets appear broadly consistent rather than systematically mispriced.